Kota Fujimoto

Project Researcher

Kyoto Institute of Technology, Kyoto, Japan

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Education

The University of Tokyo — *Tokyo, Japan*

Ph.D. in Electrical Engineering and Information Systems — *2026*

- Advisor: Dr. Hiroshi Fujimoto

The University of Tokyo — *Tokyo, Japan*

M.S. in Advanced Energy — *2023*

- Advisor: Dr. Hiroshi Fujimoto

The University of Tokyo — *Tokyo, Japan*

B.E. in Aeronautics and Astronautics — *2021*

- Advisor: Dr. Koji Uenishi

Appointments

KYOTO Design Lab, Center for the Possible Futures, Kyoto Institute of Technology — *Kyoto, Japan*

Project Researcher — *Apr. 2026 – present*

Ishizawa Lab, Institute of Industrial Science, The University of Tokyo — *Tokyo, Japan*

Visiting Researcher — *Apr. 2026 – present*

e-Mobility and Control Lab, Graduate School of Frontier Sciences, The University of Tokyo — *Tokyo, Japan*

Visiting Researcher — *Apr. 2026 – present*

Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX), The University of Tokyo — *Tokyo, Japan*

Project Student — *Apr. 2023 – Mar. 2026*

Publications

Published

Kota Fujimoto, Hiroshi Fujimoto, "Beat Switching Receiving Power Control Focusing on Vehicle-Side Circuit Dynamics for In-flight Inductive Power Transfer Systems of UAVs," *IEEJ Journal of Industry Applications*, Advance online publication, Oct. 2026, doi: 10.1541/ieejjia.20260133.

Yifan Xu, Clara Cheung, Ming Shan Ng, Akilu Yunusa Kaltungo, Tsukasa Ishizawa, **Kota Fujimoto**, "Construction robotics: a systematic review of robot types, applications and human robot collaborations," *Results in Engineering*, Vol. 30, 110406, Jun. 2026, doi: 10.1016/j.rineng.2026.110406.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive Machine Tool Based on Mode Decoupling With Virtual Viscosity," *IEEE/ASME Transactions on Mechatronics*, Vol. 30, No. 4, pp. 3054-3062, Aug. 2025, doi: 10.1109/TMECH.2025.3567273.

Takuma Katagiri, Kentaro Yokota, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Wind Tunnel Verification of Velocity Control in Wing Coordinate System using Acceleration-Based Disturbance Observer for Tilt-Wing eVTOL," *IEEJ Journal of Industry Applications*, Vol. 14, No. 2, pp. 188-195, Mar. 2025, doi: 10.1541/ieejjia.24005235.

Conference and Presentation

Peer-reviewed

Yifan Xu, Clara Cheung, Ming Shan Ng, Akilu Yunusa Kaltungo, Tsukasa Ishizawa, **Kota Fujimoto**, "Analysing Construction Robotics Safety Standards for Safety Modelling and Simulation," *in 43rd International Symposium on Automation and Robotics in Construction (ISARC)*, pp. 810-817, Singapore, Jun. 2026, doi: 10.22260/ISARC2026/0104.

Ami Kumaki, **Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Masataka Sakamoto, Yuki Terada, "Jerk-Timing-Based Polynomial Trajectory Optimization for Vibration Suppression in Machine Tool Base," *in 2026 IEEE 19th International Conference on Advanced Motion Control (AMC)*, pp. 1-6, Daegu, Korea, Mar. 2026, doi: 10.1109/AMC67705.2026.11435830.

Yusei Fujishima, **Kota Fujimoto**, Hiroshi Fujimoto, Yoshihiro Isaoka, Masataka Sakamoto, Yuki Terada, "Modeling and Analysis of Coupled System for Circular Motion in Machine Tools with XY-Axis Stage," *in 2026 IEEE 19th International Conference on Advanced Motion Control (AMC)*, pp. 1-6, Daegu, Korea, Mar. 2026, doi: 10.1109/AMC67705.2026.11435863.

Yuki Tanaka, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Altitude Optimization above Transmitter Coil for In-flight Inductive Power Transfer Using Dynamic Programming," *in 2026 IEEE 19th International Conference on Advanced Motion Control (AMC)*, pp. 1-6, Daegu, Korea, Mar. 2026, doi: 10.1109/AMC67705.2026.11435825.

Mayuko Hirota, Manto Kamiya, Yuto Naoki, **Kota Fujimoto**, Kentaro Yokota, Binh-Minh Nguyen, Sakahisa Nagai, Osamu Shimizu, Hiroshi Fujimoto, "Inertia Estimation About z-Axis Using Motor Torque for Multi-Rotors," *in The 12th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Daegu, Korea, Mar. 2026.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "Practical Implementation of High Precision Control Based on Mode Decoupling with Virtual Viscosity for Parallel Twin-Drive Machine Tool System," *in The 12th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Daegu, Korea, Mar. 2026.

Yutaka Shikauchi, **Kota Fujimoto**, Osamu Shimizu, Hiroshi Fujimoto, Shuji Kawano, "Verification of Big Data Analysis on Dynamic Wireless Power Transfer for Electric Vehicles Focused on Traffic Signal GPS," *in The 7th International Electric Vehicle Technology Conference (EVTeC)*, pp. 1-5, Yokohama, Japan, May 2025.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Parallel Twin-Drive Machine Tool System Based on Offline Parameter Identification Focusing on Differential Mode," *in The 11th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Matsue, Japan, Mar. 2025.

Takuma Katagiri, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Trajectory Tracking Control of Tilt-Wing eVTOL Using Aerodynamic Coupling Observer and Wing Angle Correction Method," *in 2025 IEEE International Conference on Mechatronics (ICM)*, pp. 1-6, Wollongong, Australia, Feb. 2025, doi: 10.1109/ICM62621.2025.10934770.

Pinzhe Feng, Yuto Naoki, **Kota Fujimoto**, Kentaro Yokota, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Short Distance Landing Control with Reverse Thrust of Variable Pitch Propeller for Fixed Wing Electric Aircraft," *in 2025 IEEE International Conference on Mechatronics (ICM)*, pp. 1-6, Wollongong, Australia, Feb. 2025, doi: 10.1109/ICM62621.2025.10934769.

Kota Fujimoto, Osamu Shimizu, Hiroshi Fujimoto, Shuji Kawano, Jin Katsuya, Tetsu Sato, "Feasibility Study for Dynamic Wireless Power Transfer in Simulation with Big Data Focusing on Battery Capacity Requirement of Electric Vehicles," *in 2024 International Symposium on Power Electronics, Electrical Drives, Automation and Motion (SPEEDAM)*, pp. 42-46, Ischia, Italy, Jun. 2024, doi: 10.1109/SPEEDAM61530.2024.10609208.

Sakahisa Nagai, Yusuke Satoh, **Kota Fujimoto**, Hiroshi Fujimoto, "Sensorless Drone Detection for In-Flight Inductive Power Transfer Using Transmitter-Side Voltage Pulses," *in 2024 IEEE Wireless Power Technology Conference and Expo (WPTCE)*, pp. 557-561, Kyoto, Japan, May 2024, doi: 10.1109/WPTCE59894.2024.10557296.

Takuma Katagiri, Kentaro Yokota, **Kota Fujimoto**, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Velocity Control in Wing Coordinate System Using Acceleration-based Disturbance Observer for Tilt-wing eVTOL," *in The 10th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Kyoto, Japan, Feb. 2024.

Shogo Yamada, **Kota Fujimoto**, Hiroshi Fujimoto, "Basic Study on Robust Perfect Tracking Control Based on Parameter Estimation Reducing Outlier Effects," *in The 10th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Kyoto, Japan, Feb. 2024.

Kota Fujimoto, Hiroshi Fujimoto, Alessandro Correa Victorino, Pedro Castillo, "Optimal Energy Trajectory Generation Based on Pitch-Dependent Mutual Inductance Model for In-Flight Inductive Power Transfer of Drones," *in 2024 IEEE 18th International Conference on Advanced Motion Control (AMC)*, pp. 1-6, Kyoto, Japan, Feb. 2024, doi: 10.1109/AMC58169.2024.10505445.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System Based on Mode Decoupling with Virtual Viscosity: Equivalent Controller Transform for Machine Tool," *in 2024 IEEE 18th International Conference on Advanced Motion Control (AMC)*, pp. 1-6, Kyoto, Japan, Feb. 2024, doi: 10.1109/AMC58169.2024.10505638.

Kota Fujimoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Test Bench Study on Attitude Estimation in Ground Effect Region Based on Motor Current for In-Flight Inductive Power Transfer of Drones," *in 49th Annual Conference of the IEEE Industrial Electronics Society (IECON)*, pp. 1-6, Singapore, Oct. 2023, doi: 10.1109/IECON51785.2023.10312055.

Yusuke Satoh, **Kota Fujimoto**, Ryo Matsumoto, Binh-Minh Nguyen, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Received Power Control of In-Flight Inductive Power Transfer for Drones by Active Rectifier Switching and Altitude Regulation," *in 49th Annual Conference of the IEEE Industrial Electronics Society (IECON)*, pp. 1-6, Singapore, Oct. 2023, doi: 10.1109/IECON51785.2023.10312000.

Kota Fujimoto, Takumi Hamada, Hiroshi Fujimoto, "Proposal on Model Based Current Overshoot Suppression of Receiver Side Coil in Drone Wireless Power Transfer System," *in 2022 Wireless Power Week (WPW)*, pp. 235-239, Bordeaux, France, Jul. 2022, doi: 10.1109/WPW54272.2022.9854039.

Non-peer-reviewed

Ami Kumaki, **Kota Fujimoto**, Hiroshi Fujimoto, Masataka Sakamoto, Yoshihiro Isaoka, Yuki Terada, "Trajectory Generation under Speed-Dependent Acceleration Constraints for Residual Vibration Suppression through Deceleration Jerk Design: Application to Machine Tools," *in IEE-Japan Technical Meeting on Industrial Instrumentation and Control (IIC)*, No. IIC-26-016, pp. 89-94, Tokyo, Japan, Sep. 2026.

Yusei Fujishima, **Kota Fujimoto**, Hiroshi Fujimoto, "Proposal of Physical-coordinate Additive Perfect Tracking Control Focusing on Rigid-body Modes in Torsional Vibration System," *in IEE-Japan Technical Meeting on Industrial Instrumentation and Control (IIC)*, No. IIC-26-011, pp. 61-66, Tokyo, Japan, Sep. 2026.

Ten Arai, Maho Matsukura, Daigo Takasaki, Isamu Moriai, Kenshin Yamakawa, Keisuke Yorita, **Kota Fujimoto**, Hiroyuki Koizumi, Kimiya Komurasaki, "Reduction of Electromagnetic Disturbances in a Wireless Power Transfer System Used in Thrust Stands for Micro Propulsion Systems," *in The 65th Conference on Aerospace Propulsion and Power*, Oita, Japan, Mar. 2026.

Mayuko Hirota, Manto Kamiya, Yuto Naoki, **Kota Fujimoto**, Binh-Minh Nguyen, Sakahisa Nagai, Osamu Shimizu, Hiroshi Fujimoto, Kentaro Yokota, "Basic Study on Z-axis Inertia Estimation Method Using Counter-torque Observer in Multirotors," *in IEE-Japan Technical Meeting on Mechatronics Control (MEC)*, No. MEC-25-007, Tokyo, Japan, Sep. 2025.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System of Machine Tool Based on Mode Decoupling with Virtual Viscosity: Feedforward Realization of Virtual Viscosity on Two-Inertia System," *in IEE-Japan Technical Meeting on Mechatronics Control (MEC)*, No. MEC-24-012, Tokyo, Japan, Sep. 2024.

Kota Fujimoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Motor Current Based Attitude Estimation Considering Ground Effect for In-flight Inductive Power Transfer of Drones: Basic Concept and Test-Bench Evaluation," *in 2024 IEE-Japan Industry Applications Society Conference (JIASC)*, No. 1-60, Mito, Japan, Aug. 2024.

Kota Fujimoto, Yusuke Satoh, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Receiving-Side System Design of In-flight Inductive Power Transfer for Drones Considering Weight Limitation," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (PE/PSE/SPC)*, No. PE-24-059,PSE-24-071,SPC-24-113, Goto, Japan, Mar. 2024.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System of Machine Tool Based on Mode Decoupling with Virtual Viscosity: Basic Study on Two-Inertia System," *in IEE-Japan Technical Meeting on Mechatronics Control (MEC)*, No. MEC-23-011, Tokyo, Japan, Sep. 2023.

Kota Fujimoto, Osamu Shimizu, Hiroshi Fujimoto, Jin Katsuya, Tetsu Sato, "Requirement Analysis for Dynamic Wireless Power Transfer with Big Data from Real Vehicles," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (PE/PSE/SPC)*, No. PE-23-084,PSE-23-090,SPC-23-140, Nago, Japan, Mar. 2023.

Yusuke Satoh, **Kota Fujimoto**, Ryo Matsumoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Basic Study on Flight Altitude and Battery Current Control in In-flight Wireless Power Transfer for Drones," *in IEE-Japan Joint Tech-*

nical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (PE/PSE/SPC), No. PE-23-085,PSE-23-091,SPC-23-141, Nago, Japan, Mar. 2023.

Kota Fujimoto, Kentaro Yokota, Sakahisa Nagai, Hiroshi Fujimoto, "Modeling and Verification of Pitch-Dependent Coupling Coefficient for WPT to Flying Drone," *in 2021 IEE-Japan Industry Applications Society Conference (JIASC)*, No. 4-8, Nagaoka, Japan, Aug. 2021.

Presentations

Kota Fujimoto, "UAV-Oriented Control and System Design for In-flight Inductive Power Transfer Systems," *in Workshop on Advanced Aerial Robotics 2nd Edition*, Tokyo, Japan, Oct. 2026. (Invited)

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "Practical Implementation of High Precision Control Based on Mode Decoupling with Virtual Viscosity for Parallel Twin-Drive Machine Tool System," *in The 12th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Daegu, Korea, Mar. 2026.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive Machine Tool Based on Mode Decoupling With Virtual Viscosity," *in IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, Hangzhou, China, Jul. 2025.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Parallel Twin-Drive Machine Tool System Based on Offline Parameter Identification Focusing on Differential Mode," *in The 11th IEEJ International Workshop on Sensing, Actuation, Motion Control, and Optimization (SAMCON)*, Matsue, Japan, Mar. 2025.

Kota Fujimoto, "Energy Maximum Trajectory Generation Focusing on Motion Dynamics for In-flight Inductive Power Transfer of Drones," *in Global Green Transformation Workshop at The University of Manchester and the University of Tokyo Joint Research Symposium*, Manchester, UK, Oct. 2024.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System of Machine Tool Based on Mode Decoupling with Virtual Viscosity: Feedforward Realization of Virtual Viscosity on Two-Inertia System," *in IEE-Japan Technical Meeting on Mechatronics Control (MEC)*, Tokyo, Japan, Sep. 2024.

Kota Fujimoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Motor Current Based Attitude Estimation Considering Ground Effect for In-flight Inductive Power Transfer of Drones: Basic Concept and Test-Bench Evaluation," *in 2024 IEE-Japan Industry Applications Society Conference (JIASC)*, Mito, Japan, Aug. 2024.

Kota Fujimoto, Osamu Shimizu, Hiroshi Fujimoto, Shuji Kawano, Jin Katsuya, Tetsu Sato, "Feasibility Study for Dynamic Wireless Power Transfer in Simulation with Big Data Focusing on Battery Capacity Requirement of Electric Vehicles," *in 2024 International Symposium on Power Electronics, Electrical Drives, Automation and Motion (SPEEDAM)*, Ischia, Italy, Jun. 2024.

Kota Fujimoto, Yusuke Satoh, Sakahisa Nagai, Hiroshi Fujimoto, "Basic Study on Receiving-Side System Design of In-flight Inductive Power Transfer for Drones Considering Weight Limitation," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (PE/PSE/SPC)*, Goto, Japan, Mar. 2024.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System Based on Mode Decoupling with Virtual Viscosity: Equivalent Controller Transform for Machine Tool," *in 2024 IEEE 18th International Conference on Advanced Motion Control (AMC)*, Kyoto, Japan, Feb. 2024.

Kota Fujimoto, Hiroshi Fujimoto, Alessandro Correa Victorino, Pedro Castillo, "Optimal Energy Trajectory Generation Based on Pitch-Dependent Mutual Inductance Model for In-Flight Inductive Power Transfer of Drones," *in 2024 IEEE 18th International Conference on Advanced Motion Control (AMC)*, Kyoto, Japan, Feb. 2024.

Kota Fujimoto, Sakahisa Nagai, Binh-Minh Nguyen, Hiroshi Fujimoto, "Test Bench Study on Attitude Estimation in Ground Effect Region Based on Motor Current for In-Flight Inductive Power Transfer of Drones," *in 49th Annual Conference of the IEEE Industrial Electronics Society (IECON)*, Singapore, Oct. 2023.

Kota Fujimoto, Hiroshi Fujimoto, Yoshihiro Isaoka, Yuki Terada, "High Precision Control for Twin-Drive System of Machine Tool Based on Mode Decoupling with Virtual Viscosity: Basic Study on Two-Inertia System," *in IEE-Japan Technical Meeting on Mechatronics Control (MEC)*, Tokyo, Japan, Sep. 2023.

Kota Fujimoto, Osamu Shimizu, Hiroshi Fujimoto, Jin Katsuya, Tetsu Sato, "Requirement Analysis for Dynamic Wireless Power Transfer with Big Data from Real Vehicles," *in IEE-Japan Joint Technical Meeting on Power Engineering/Power Systems Engineering/Semiconductor Power Converter (PE/PSE/SPC)*, Nago, Japan, Mar. 2023.

Kota Fujimoto, Takumi Hamada, Hiroshi Fujimoto, "Proposal on Model Based Current Overshoot Suppression of Receiver Side Coil in Drone Wireless Power Transfer System," *in 2022 Wireless Power Week (WPW)*, Bordeaux, France, Jul. 2022.

Kota Fujimoto, Kentaro Yokota, Sakahisa Nagai, Hiroshi Fujimoto, "Modeling and Verification of Pitch-Dependent Coupling Coefficient for WPT to Flying Drone," *in 2021 IEE-Japan Industry Applications Society Conference (JIASC)*, Nagaoka, Japan, Aug. 2021.

Awards, Fellowships, & Grants

Awards

Aug. 2025 **2024 IEEJ Industry Applications Society Excellent Presentation Award**, The Institute of Electrical Engineers of Japan

2022 **APET Excellent Poster Presentation Award 3rd Prize**, Alliance for Advanced Power & Environmental Technology

Aug. 2022 **2021 IEEJ Excellent Presentation Award**, The Institute of Electrical Engineers of Japan

Fellowship

Apr. 2023 – Mar. 2026 **Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)**, The University of Tokyo

Grant

2025 **The Dispatch for International Meetings Program (for IEEE AIM2025)**, Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)

2024 **The Dispatch for International Meetings Program (for Global Green Transformation Workshop at The University of Manchester and the University of Tokyo Joint Research Symposium)**, Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)

2024 **The Dispatch for International Meetings Program (for SPEEDAM 2024)**, Fostering Advanced Human Resources to Lead Green Transformation (SPRING GX)

2023 **Overseas Travel Expense Assistance**, The Telecommunications Advancement Foundation

Sep. 2023 – Aug. 2024 **La Bourse de Mobilité Internationale de Sorbonne Universités**, Sorbonne Université

2022 **Researcher Overseas Training Grant**, Suzuki Foundation

Teaching/Supervising Experience

2021 – 2025 **Graduate Teaching Assistant**, Department of Electrical and Electronic Engineering, Faculty of Engineering, The University of Tokyo

Outreach & Professional Development

Service and Outreach

2022 **Workshop in Kashiwanoha innovation festival**, Operational Staff at Craft Construction Course of Dynamic Wireless Power Transfer — *Kashiwa, Japan*

Internship

May – Aug. 2024 **Heudiasyc Lab, Université de technologie de Compiègne**, Project internship student — *Compiègne, France*

Aug. – Nov. 2023 **Heudiasyc Lab, Université de technologie de Compiègne**, Project internship student — *Compiègne, France*

2022 **R&D Mechatronics team**, ACSL Ltd. — *Tokyo, Japan*

Professional Memberships

The Institute of Electrical and Electronics Engineers, Incorporated Member
The Institute of Electrical Engineers of Japan Member
The Japan Society for Aeronautical and Space Sciences Member
Society of Automotive Engineers of Japan, Inc. Member